Less work, more care

Automating sports memory-matching using recommender systems

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Abstract

With the ageing of the population in many countries, the number of people living with dementia will increase, creating a bigger need for ways to improve the well-being of these people and reduce social isolation. Sports reminiscence therapy, which invites participants to revisit memorable sporting moments through photographs, has proven effective but remains labour-intensive because each picture must be hand-matched to an individual's life story. This thesis explores techniques that can automate that matching process and thus widen access to the therapy.

We assemble an already rated dataset by our external collaborator "Sports & Memory" that links rich personal profiles (consisting of demographic details, sporting interests and key life events) to a selected set of sports memory cards (*terugblikken.com*). Three modelling tracks are compared. First, cosine-similarity baselines measure textual overlap between profiles and card descriptions. Second, "simple" machine learning pairs diverse text encoders with several regressors, (linear, tree-based and distance-based learners). Third, fusion models add visual embeddings from pre-trained image encoders to the textual features.

To accurately match a personal profile of people with dementia to the most relevant sports image, we have tried both memory-based approach (cosine similarity) and model-based approach (simple textual machine learning & machine learning with fusion models) with four different encoders (3 LLMs & TF-IDF). We found that, for cosine similarity, choosing a larger LLM with more comprehensive parameters performed better in our recommendation task. For the two model-based approaches, choosing an LLM with a smaller model gave better performance. Adding a visual layer worsens the performance of the model. Compared to TF-IDF, using an LLM to make semantic embeddings showed higher effectiveness of the learning process and the prevention of potential overfitting.

Contents

1. Introduction

One of the biggest challenges many (Western) countries will face in the next decades, is the ageing of the population. On a global scale, the share of people over sixty years of age will double in the 2016-2050 period. Where ten years ago only 11% of the global population were aged 60+, this percentage is projected to go up to 22% by 2050 (Kanasi, Ayilavarapu, & Jones, 2016). When looking at the European Union, we see that the share of people over 65 years of age is already over one fifth of the population: 21.6% (Eurostat, 2025). The biggest difference, compared to the last century, is visible in the age group of 75–84 year-olds. From the start of the 21st century until its midway point, this group is expected to grow by 56.1% (Eurostat, 2024). The demographic shift towards an older population is likely to bring with it many challenges.

These challenges can be societal, like the pressure on the health care system with a relatively high share of people that require care and an increasingly small group of working age people that could provide this care. This could lead to an intensified pressure on the people in the working age, who need to pay the taxes to cover an increasingly high national health care bill (Rechel, et al., 2013). However, the challenge is not only in 'keeping the country running', but also very much in giving an increasingly large group of elderly the care and support they require. There are many aspects to this complex problem, but in this study we will focus on the growing group of elderly with dementia - and the problems they face.

Given that the population in many Western countries is aging at a rapid pace, and that dementia is a disease that becomes more prevalent with age, it is no surprise that the number of people with dementia is increasing dramatically (Winblad, P., et al., 2010). In the Netherlands, the target country of this research, there are currently 280,000 people with dementia – a number that might double by 2040, according to the Dutch government (Rijksoverheid, 2024).

People with dementia experience a lot of negative effects of the disease on their (psychological) well-being. The quality of life goes down as they experience poorer physical health, struggle with their social relationships, and are more likely to experience depression (Martyr, et al., 2018). It is therefore of great importance that civilians know how to help people with dementia – and how to prevent them (as much as possible) from experiencing the possible negative consequences of a life with dementia. To do this, much research has been conducted, and many therapies and guidelines have been developed. In this research, we will be looking at one specific therapy: reminiscence therapy.

Reminiscence is defined as: "a free-flowing process of thinking or talking about one's experiences in order to reflect on and recapture significant events of a lifetime." (Latha, et al., 2014). In this master thesis, we will report the findings of our research in the subgenre of sports reminiscence therapy - in cooperation with the Sport & Memory initiative, which is targeting Dutch people with dementia (Sport & Memory, 2025). This type of therapy can be performed in many different ways, but we will be focussing on the use of photographs as a tool for reminiscence. By using these photographs, combined with some information about the person or setting in the photograph, the person with dementia will be guided in the process of reliving sports memories from earlier in their life. For this process to be as meaningful and successful as possible, the photographs that are used need to be relevant to the person in question. This process of matching a person to the most relevant pictures is quite labour-intensive, which limits the number of people that can be included in the therapy.

In this research, we will look at how the process of matching a person to the most relevant photograph could be done in the most optimal and efficient way, to increase the potential number of people that could get this therapy. For this matching process, we have explored different ways in which large language models (LLMs) could be used. LLMs, which are very large deep-learning models that are trained on a vast amount of data, are a popular tool to perform matching processes of many kinds, including healthcare settings in which a patient is matched to the best clinical trial (Jin, et al., 2024). A broad range of different LLM models exist, all of them with their own strengths and weaknesses, which will make some models more useful in specific situations than other models.

We will be testing different models that could have the potential to perform well in this specific situation. These models will be adjusted to the specific characteristics of the project. To match a person to the most relevant photograph from a large database, many different factors play a role. These could be characteristics of the photograph or of the person in question. Think about characteristics like gender, age, birthplace, and favourite sport. Taking all relevant factors into consideration, the model will be modified to these specific tasks, in order to make the matching process as efficient as possible.

In this study, we will be aiming to answer the following research question:

How can we accurately automate the process of matching personal profiles of people with dementia to the most relevant sports images for sports reminiscence?

2. Literature review

In the literature review, relevant insights about the topic of this research will be discussed by looking at the existing literature. In doing so, relevant concepts will be explained, and context will be provided about the academic landscape this study finds itself in - in order to show the academic and societal relevance of this study.

2.1. Reminiscence therapy

The power of reminiscence therapy has been subject to research for over half a century. Papers and books about the effect of helping elderly remember scenes from the past were published as far back as the 70s (Kaminsky, 1979). This can be done using tangible prompts such as music, photographs or videos, which will make it easier to bring back memories. These methods can be used as a form of therapy, where the therapist/caretaker can use reminiscence in a range of settings using a variety of formats (Woods, et al., 2018; Lazar, Thompson, & Demiris, 2014).

Reminiscence therapy is usually targeting elderly – the goal is to improve (psychological) well-being of these people by, for instance, reducing depression, loneliness, anxiety, or a combination of those (Elias, Neville, & Scott, 2015). There is a vast amount of evidence that reminiscence therapy can be very effective and can have positive effects on the participants. Positive effects that have been reported are: improved mood, higher self-esteem, more frequent and more intense feelings of happiness, improvement in specific cognitive functions, enhanced emotional balance, increased sense of meaning, and reduced feelings of depression, loneliness, and anxiety (Subramaniam & Woods, 2012; Hsieh & Wang, 2003; Tam, et al., 2021; Chen, Shapie, & Li, 2024).

Most of the studies on reminiscence therapy target elderly people that suffer from dementia, usually living in care centres. As these people are having problems with bringing back memories, the impact of a well-performed reminiscence therapy session can be significant for them (Subramaniam & Woods, 2012). Over the years, a lot of initiatives have taken place in care centres for elderly with dementia, which resulted in a vast corpus of literature about the topic. In the last decades, more research has been conducted on the use of specific types of memories, leading to different branches in reminiscence therapy.

2.2. Sports reminiscence therapy

One of those is the branch of sports memories. Sports play a major role in the life of many people – both playing and watching sports can trigger strong emotions and sports play a central role in the social life of many (Peterson, 2014). Bringing back these moments through reminiscences can trigger similar emotions and therefore make people re-experience moments that were dear to them (McDermott, 2018; Watson, Parker, & Swain, 2018). Over the last two decades, a lot of sports reminiscence therapy projects have started. Usually there are external therapists or researchers involved in the implementation of such projects, though the possibility of training care home workers to give these workshops is being explored in recent years (Clark, et al., 2017). The potential role that other organisations where elderly suffering from dementia gather could play, like Christian or other religious organisations, has been the interest of recent research as well (Watson, Parker, & Swain, 2018).

In some of these projects, there is a focus on just football-related memories. As this is the biggest sport in Europe, there is more material for football than for any other sport. Also, it is more likely that the people receiving the therapy will be interested in football, rather than any other sport. Especially in the United Kingdom, football reminiscence therapy has been a heavily researched topic, leading to a vast network of initiatives on this topic all over the UK (Football Memories, 2025; McDermott, 2018; Sport England, 2017; Exeter University, 2024). Most of the sports-related reminiscence therapy research targets men, as they tend to have a higher emotional connection with sports (Tolson & Schofield, 2012). Some of these look specifically at the role of perceived masculinity (Sass, Surr, & Lozano-Sufrategui, 2021). As a result of this gender-disparity in the research done on this topic, more research has been targeting female elderly in recent years (Oatley, 2021).

2.3. People with dementia in the digital age

We are living in a time of non-stop expansion of technological possibilities, in which society is increasingly digital. This offers great opportunities in many aspects but also brings great challenges. In general, the digital literacy of people with dementia is quite low, which can be problematic in the current day and age (Zaid, et al., 2022). In the last decade(s), a lot of research has been conducted on how to improve the digital literacy of this group (and of elderly in general). Because even though these emerging new technical developments might seem problematic for this group, they could also offer great solutions for many problems that people with dementia face. As of now, it is frequently assumed that people with dementia are inherently digitally illiterate – and that this is something that can't change (Fletcher & Brown, 2025). However, many studies indicate otherwise.

There are two approaches to this: trying to make the people with dementia more digitally literate through training and guidance or making the digital environment more suitable for people with dementia and lower digital literacy. Both of these have gotten a major boost during covid, where everyone suddenly got a lot more dependent on technology to keep in touch with other people and sometimes even to receive treatment. Both approaches have shown potential for improving the quality of life of people with dementia and reducing isolation/loneliness in this group (Rai, et al., 2022). A side note will have to be placed about that the digital literacy training for people with severe forms of dementia is not likely to be very effective, and more potential seems to be seen in using technology in a more user-friendly way for people with dementia (Martínez-Alcalá, et al., 2021). Examples of this are the implementation of more systems that work with voice-command, or the use of Large Language Models to e.g., reduce cognitive demand by simplifying texts or overcoming word-finding problems (Fletcher & Brown, 2025).

Something that needs to be mentioned is that implementing these techniques could enlarge the inequalities in dementia care, especially for marginalised groups, as they are less likely to receive good care and guidance in the digital world. Furthermore, ethical concerns about privacy and autonomy arise when the environment of the people with dementia is increasingly digitized – think of smart home technologies (Giebel, 2023). Nevertheless, modern technology has great potential to change dementia care for the better – resulting in a hybrid form of care, which could lower the pressure on the health care system.

Although the topic of reminiscence therapy and people with dementia has already been studied by numerous researchers, it is highly noticeable that our research question, if only reflecting the research method from existing literature, could only be answered by qualitative research consisting of interviews, field study or a longitudinal study. As researchers with an applied data science background, we have noticed it as a great opportunity to address the research gap by offering a quantitative solution using statistical and data science methods and evaluative metrics. We plan to explore the possibility of using a recommender system to strictly target people with dementia with products designed for the purpose of reminiscence therapy. We expect our research to assume the role of a valuable and scientifically relevant complement to the existing qualitative research repertoire of people with dementia and their relationship with technology, given the quantitative methods and machine learning techniques inside the establishment of a recommender system (Roy & Dutta, 2022). Moreover, we expect our research to be practically and socially relevant as well, provided that a successful recommender system is performing well in terms of "efficiency", "effectiveness", "enjoyment" and "trustworthy", which helps the process of assigning reminiscence-proned items to people with dementia to achieve a more efficient, powerful and positive therapeutic effect (Cheung et al, 2019, p.7-14). Given the overarching nature of our research, it is necessary to provide below an overview of existing recommender system studies for a more comprehensive reference, before delving deeper into our own research - which aims to address people with dementia with the goal of reminiscence therapy by using the method of recommender system.

2.4. Recommender Systems

For a general definition, a recommender system is one of the "information filtering tools, offering users suitable personalized information and content" (Roy & Dutta, 2022, p1-2). The whole system could be understood as three parts interconnected with each other: a set of users with their personal information, a set of items with product information, and a utility function testing the usefulness of items to users (Burke et al., 2011, p.14). The research and development of recommender systems have already started in the late 20th century and continued for the following decades, which engendered abundant practices and significant progress in this field (Wang et al., 2024, p.4). Here, a general introduction to the field of recommender systems regarding the variation of its algorithmic types and industrial sectors is briefly delineated below.

2.5. Recommender systems and their algorithmic types

Most literature have mentioned the following three algorithms as the main types of recommender systems: Content-based filtering, collaborative filtering and hybrid filtering (Ameen, 2019; Blut et al., 2023; Cheung et al., 2019; Kurek et al., 2024; Roy & Dutta, 2022). Other than those three, knowledge-based filtering has also been listed by some researchers as the fourth one among main recommender algorithms (Ameen, 2019; Cheung et al., 2019; Kurek et al., 2024). Occasionally, one could also find some additional algorithmic types such as "interactive RA" or "self-serving recommendations" being mentioned in some research, but a broad existence of them across a wide scope of different recommender system literature was unfortunately not established (Blut et al., 2023, p.442). Therefore, only the four main algorithmic types being widely mentioned are covered in more detail here.

For a brief explanation, content-based filtering focuses on product information and signifies a mechanism to group items with high similarity under the same set. For any user who has shown sufficient preference to items under that set, he would then be recommended with other items from the same group (Roy & Dutta, 2022, p.3). Collaborative filtering, on the other hand, concentrates on user information and is designed to group similar users under the same set. For any product having received high ratings from users under that set, it would then be recommended to other users from the same group (Roy & Dutta, 2022, p.4). These two methods have already been conducted at the earlier stage of recommender system research and practice, and remain widely applicable nowadays. However, neither of them could be exempted from obvious shortcomings. For both content-based and collaborative filtering, a cold start problem could be the most prominent disadvantage: Recommendation struggles when new items without sufficient metadata, users' histories or user-product interactions such as ratings or clicking are introduced into the system (Blut et al., 2023, p.442). In addition, collaborative filtering could encounter the problem of data sparsity when users only rate very few items in the item set while content-based filtering could suffer from limitation to expand the users' interest scope (Blut et al., 2023, p.442).

In order to maintain the advantages while addressing the shortcomings of the two algorithmic methods above, hybrid filtering was proposed in the later stage and has developed into different methodological directions such as "feature combination", "feature augmentation", "weighted hybridation", "mixed hybridation" and "switch hybridation", etc. (Roy & Dutta, 2022, p.6). For a brief explanation, hybrid filtering "can use content-based filtering in a collaborative method or a collaborative-filtering technique in a content-based method" (Roy & Dutta, 2022, p.6). Nonetheless, hybrid filtering could not be fully exempted from the cold start problem either if an initial user-item interaction is missing. In this case, knowledge-based filtering would become the better choice, which, based on its definition, relies heavily on expert knowledge, individual case, defined rules and constraints rule-based personalization (Ameen, 2019).

2.6. Recommender systems and the studied sectors

As a continuous development over dozens of years, the research or practice of recommender systems have already been in different sectors (Wang et al., 2024, p.4). According to the meta-analysis of Roy & Dutta (2022, p.10), some of the most prolific sectors include "entertainment", "social media/others" and "web/e-commerce", which respectively accounts for 26%, 20% and 17% among retrieved articles about recommender systems. In comparison, the health sector reached 13%, demonstrating as a relatively modest field with strong potential for further research (2022, p.10). Furthermore, among existing recommender system research in the health sector, most of them choose to address the general public, healthy adults or patients with specific physiological complaints, such as back pain, overweight or diabetes (Cheung et al., 2019, p.5-6). They have the research objective of general health promotion, prevention, physiological nuisance relief, such as weight loss, or behavioral nuisance relief, such as smoke cessation (Cheung et al., 2019, p.5-6). In contrast, people with dementia and the objective of emotional and, potentially speaking, cognitive betterment have been largely neglected in the extant research repertoire of recommender systems in the health sector. In a relatively isolated case, Rao et al. (2021) has paid specific attention to people with dementia and studied how well personalized music could help recall memories about the songs in their young age for people with dementia. Nonetheless, their research was conducted methodologically with qualitative survey and no recommender system has been applied in the process.

2.7. The recommender system pipeline

The pipeline of a recommender system generally follows five consecutive steps: Data collection, feature engineering, feature encoder, ranking function and recommendation (Wang et al., 2024, p.3). The methodology applied in recommender systems shows variation in every step of this pipeline, yet overall, most of these variations could be loosely categorized into two main methods: memory-based approach and model-based approach (Roy & Dutta, 2022, p.4-6). Both of them follow the five consecutive steps above.

For a general explanation, the memory-based approach receives users' ratings of products and transforms them into a single utility matrix matching different item ratings to each user as the first step. Then, it passes a function to calculate the user-user or item-item similarity matrix. One of the most widely used functions in relevant literature for this process is cosine similarity, which has the dot product of vector embeddings of users information and those of item information divided by the product of each vector's magnitude (Colangelo et al, 2025; Johari & Laksito, 2021; Kurek et al., 2024; Yunanda et al., 2022). Before applying cosine similarity, textual information about the users and items need to be carefully accumulated and compiled, which in many cases entails meticulous data collection and feature engineering such as keyword extraction based on professional or expert knowledge (Colangelo et al, 2025). To conduct a successful pipeline in memory-based approach, a utility matrix is always necessary from the beginning to the end and the computation burden could be easily overwhelmed for larger datasets.

Model-based approaches, on the other hand, apply machine learning algorithms to interpret the patterns connecting user data and item data (Roy & Dutta, 2022, p.5). It uses a machine learning function to train and test on the data and makes recommendations for new data without the need of a utility matrix, which distinguishes significantly from memory-based approach and could better tackle the cold start problem as the consequence. Besides, a model-based approach performs better when scalability of incoming data is highly valued. However, it takes longer in the training process and is more difficult to interpret the result compared to the straightforward similarity matrix and scores in a memory-based approach. According to the meta-analysis from Roy & Dutta (2022, p.19), models such as K-means clustering and fuzzy logic were used in more studies than other models such as linear SVM or random forest, but no explanation was given to address the choice discrepancy directly in this study.

Through the pipelines of memory-based approach and model-based approach, an encoding method needs to be carefully selected in both cases to numerically represent the features. In the early dates of recommender system studies,traditional encoding methods include one-hot encoding or TF-IDF (Castells & Jannach, 2023, p.19). TF-IDF is a method to weight different terms in a text and is widely used for tasks such as textual classification or clustering. It is calculated as the product of the frequency of a term in a document, the frequency of the same term in the whole corpus and a normalization part addressing document length (Sammut & Webb, 2011). In recent years, the use of Large Language Models (LLMs) has received increasing attention in the recommender system field due to its ability of understanding the semantic meanings, context-sensitive information and nuanced user preferences (Wang et al., 2024). A Large Language Model (LLM) is defined as a model trained on extensive textual data and is able to process advanced contextual meaning (Wang et al, 2024, p.7). Research has been conducted to compare the performance of different LLMs in recommending tasks (Colangelo et al., 2025; Kurek et al., 2024; Wang et al., 2024), most of which is relatively recent and an extensive discussion on this topic is yet to be achieved. Moreover, according to the result of Epure (2024), TF-IDF still manages to outperform some LLMs in specific recommendation practice.

In the next chapter, we will describe the pipeline that will be used in this study, in order to create a recommender system that is able to generate predicted relevancy scores for every participant-picture combination.

3. Data and methods

In this chapter, we will describe and show the data sources that were used in the study. Furthermore, we will explain what methodological choices were made, based on the literature review that was performed. All methods and models that were used will be described, following the same steps as we will follow in the pipeline we are using in creating a model to automate the matching process between people with dementia and the most relevant pictures.

3.1. Step 1: collecting data

Building on the basis of existing studies, we approach our research question by introducing a recommender system to match the most relevant sports image to people with dementia. The methodology follows the widely accepted five consecutive steps in recommender system literature. Firstly, we have gathered the data: user profiles, product samples and descriptions and relevancy scores (assigned by an expert) for every participant-picture combination. Given the General Data Protection Regulation's (GDPR) requirement of explicit consent for sensitive personal information and the complexity of contacting people with dementia or their caretakers for such consent, using expert-rated data was seen as a logical choice to balance the data accuracy, privacy concern and research efficiency.

3.1.1. Data collection:

As for any research, data collection is one of the most important steps. The data we used were separated into data about the participant and data regarding the products the company is making (i.e., reminiscence cards). Two types of participant data were used. The first one being data about future participants that we would generate recommendations for - i.e., inference of the recommendations. This data set includes demographic, biographical, emotional, and interest-related information about each candidate; More specifically:

- **Name**: full name of the candidate.
- **Age**
- **Residence**: current place of residence of the candidate.
- **Marital Status**: current marital status of the candidate.
- **Living Situation**: information about the current living situation of the candidate.
- **Health**: current dementia state, other information about physical status of the candidate.
- **Life Story Highlights**: Information such as former profession, hobbies, place of birth.
- **Sport Interests**: Information about the sports the candidate is/was interested in, favourite team or athletes, sport events that the candidate has attended that have been set in his/her memories.
- **Reminiscence triggers**: Sport events that have triggered the candidate's memories.
- **Preferred Content Style**: preferred style of content presentation e.g. tone of speaking, preference for type of prompts.

Alongside this, we had a tabular data set in which every participant received a relevancy score for every card that would be used. These scores were assigned by an expert from the Sport and Memory company. This data set, including the relevancy scores would be used to train and test the models on.

- **Profile columns:** resident identifier, age, gender, year and place of birth, region lived in during early adulthood, up to three stated sports interests, cognitive-status category, and a free-text remarks field.
- **Rating columns:** numeric matching scores (0–10) that an expert assigned to specific sport-memory items in combination with a participant.

As for the product, each one of the 20 items in the *terugblikken.com* collection provided by "Sports & Memory" is a double-sided, postcard-sized memory card designed to evoke sporting memories.

The front side (Figure 1) is a high-contrast photograph of either an iconic (action) shot of an athlete or an image capturing a significant sports event from the past inside a slightly tilted, polaroid-style white frame. The frame is located in a solid blue background that makes the image stand out, making it easier to be read by the candidates. Above the picture, the athlete's name or the event title appears in smaller white letters, while the surname or main descriptor is set beneath it in bold, larger capitals, establishing a clear visualization.

The back side (Figure 2) retains the same blue backdrop. A slim subheading labeled "SPORT" is followed by the specific discipline in bold (e.g., Voetbal, Wielrennen, Turnen - Dutch for football, Cycling, Gymnastics). Below that, a concise Dutch text block provides context—summarising a player's career highlights, a record-setting performance, or key facts about the featured event. The copy is left-aligned and brief enough to be read aloud easily during reminiscence sessions.

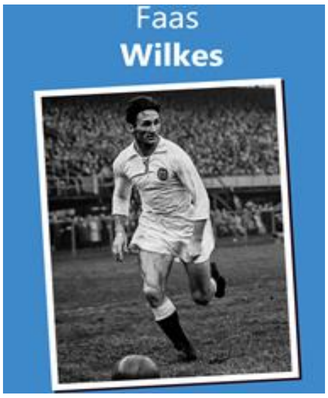

Figure 1: Card front

Figure 2: Card back

3.2. Step 2: Feature engineering

After gathering the data, we have performed some feature engineering work that we would incorporate in the models. This helps test our expectation that certain user features play a more important role than others in building a reliable recommender system.

For a general explanation, feature engineering is defined as an attempt to exploit domain knowledge to the dataset based on the distinguishing features that help the algorithm better analyze and solve the problem (Jalal, 2016, p.3). Common feature engineering techniques include univariate feature engineering, multivariate feature engineering, and domain-specific feature engineering (Verdonck et al., 2024, p.3919-3921).

In our research, we have applied the domain specific feature engineering technique by marking indicator features based on expert knowledge. Among the 10 features our user data contains, we have received the information from the external expert that age congruence between users and items is an important indicator for users' ratings on products. It was suggested by the Sport and Memory experts that if the corresponding year of the product falls into the year range of the user when he was between 15 to 30 years old, this product should receive additional weights in the recommender system. The theory behind this choice is the reminiscence bump, which indicates that the most robust and unforgettable memory from people normally appears between 10-30 years of age (Munawar et al., 2018).

Furthermore, we have assigned an additional weight when the location of a user matches the corresponding location of an item. This decision is based on the positive geographical influence found in some recommender system studies (Ye et al., 2011). A grid search between additional weights of 0.001 and 0.005 for location and between 0.005 and 0.01 for age has been adopted for the cosine similarity technique. The discrepancy between those was created on the basis of the external expert knowledge that age condition has a bigger effect. The lower and upper bounds of 0.001 and 0.01 were chosen due to the overall number of 10 features and the range of cosine similarity scores between -1 and 1: We restricted the maximal influence of any single feature and prevented disproportionate skewing of similarity scores. The supervised recommending result from the best performing encoder based on engineered features from cosine similarity would be compared to the result from the same encoder based on original feature information. If the cosine similarity result with additional weights is not any better than the original one.

3.3. Step 3: Feature encoding

For the third step, we encode user and product features with both TF-IDF and three different LLMs as a reflection of the newest trend of incorporating LLMs with recommender systems and the comparison between the performances of different LLMs and between LLM and classical TF-IDF.

Feature encoding is the step to transform features into numeric values in order to pass them into the recommender algorithm. In the feature encoding process of our research, raw personas profiles were converted into machine learning readable vectors. We did that in three parallel ways:

1. **One-hot encoding:** One-hot encoding (also known as 1-of-K or dummy coding) transforms a nominal categorical variable taking one of K mutually exclusive values into the Kronecker-delta basis $\mathbb{R}$K: it represents each sample as a K-dimensional indicator vector having one coordinate equal to 1 and the other K−1 coordinates equal to 0 (Bishop, 2006). Because these basis vectors are equidistant and pairwise orthogonal, the transformation does not impose any artificial ordering; linear models, kernel machines and neural networks therefore treat categories as qualitatively distinct and not points on some arbitrary numerical scale, fulfilling the statistical necessity that "qualitative inputs be represented by numeric or dummy coding" (Hastie et al., 2009).
2. **TF-IDF:** (*term frequency* × *inverse document frequency*) is a lexical weighting technique that elevates words appearing frequently within a particular document yet seldom across the broader corpus. By spotlighting vocabulary that most characterises a text and suppressing ubiquitous "stop-words," it concretises the intuition that distinctive language conveys meaning (Salton & Buckley, 1988). This approach became the standard conduit between raw language and statistical analysis because it preserves that qualitative insight while situating every document inside a rigorously defined vector space (Manning, Raghavan & Schütze, 2009). Subsequent scholarship interprets the same weighting as a direct measure of information content, showing that TF-IDF quantifies the uncertainty removed when a word is observed—an information-theoretic rationale for its enduring value (Aizawa, 2003).

3. **LLM sentence embeddings:** Large-language-model (LLM) sentence embeddings compress the full meaning of a clause or sentence into a single, dense vector (typically a few-hundred dimensions) so that semantically similar sentences occupy neighbouring points in the resulting space. They are produced by passing the tokenized sentence through a transformer that has been *pre-trained* on vast text corpora; the model's self-attention layers integrate left- and right-hand context, thereby encoding grammatical structure, disambiguating polysemy, and capturing pragmatic cues that bag-of-words methods discard (Devlin et al., 2018). More specifically we used three different LLM models:

    a) **All-mpnet-base-v2:** This model is built on MPNet, which merges the advantages of BERT's masked-token prediction with XLNet's permuted-token training while correcting each method's positional biases (Song et al., 2020). We include it because it offers state-of-the-art semantic performance while remaining "base"-sized, which keeps inference latency within practical limits for an interactive recommender.
    b) **MiniLm:** MiniLM compresses the self-attention patterns of a large teacher transformer into a 33 M-parameter student while preserving most of the teacher's discriminative power (the smaller "student" network is trained to mimic the attention maps and hidden states produced by the much larger "teacher," so it inherits the teacher's semantic knowledge while remaining far lighter in size) (Wang et al., 2020). The multilingual paraphrase variant maps text from 50 + languages into a shared 384-dimensional space and performs well on cross-lingual retrieval and clustering tasks. Its small footprint and broad language coverage make it ideal for our research, since we have to do with Dutch care facilities, where biographies include Dutch & English parts and computing resources are limited.
    c) **Instructor XL**: INSTRUCTOR reframes embedding generation as an instruction-following problem: each text is paired with a natural-language prompt that specifies the task or domain, and the model learns to align its vector space accordingly (Su et al., 2023). The XL version (~1.5 billion parameters) currently tops the Massive Text Embedding Benchmark, scoring about 3–4 percentage points higher than earlier baselines without any extra fine-tuning once a natural-language instruction is provided.

3.4. Step 4: Adopting ranking functions

For the fourth step, we have applied three different methods for ranking functions: One recalls the memory-based approach by calculating the cosine similarity between users and items and then testing it against the expert-labelled ratings. The other two methods recall the model-based approach by using different machine learning models on a separated training set and test set. This helps achieve a more general, comprehensive view on the performance of our recommender system and offers us an opportunity to compare the results across these two main methods in recommender system studies. Moreover, these three methods have a close connection to the traditional hybrid filtering methods, but do not fall fully into its frame given the supervised testing against the labelled ratings and the supervised learning techniques. This helps bring additional insights to the domain that has been dominated with unsupervised evaluation which typically rely on deployment and real-time feedback from new users. The first method is using cosine similarity between the profiles and product descriptions, the other two are using LLM embeddings of the profiles - where the last method adds a visual embedding of the picture itself on top of this.

3.4.1. Method 1: Cosine similarity

The memory-based approach was addressed by calculating the cosine similarity between user profiles and product descriptions. Cosine similarity was calculated with the vector of user information and that of product descriptions. Both vectors were the result of feature encoders we have chosen and presented in the section above. Mathematically speaking, cosine similarity has the dot product of vector embeddings of users information and those of product information divided by the product of each vector's magnitude. In this case, each user had a corresponding similarity score to each product between -1 and 1. From 0 to 1, the closer the similarity score to 1, the higher the similarity was between the selected user and product. From 0 to -1, the closer the similarity score to -1, the higher the opposition was between the selected user and product. The score 0 indicated the similarity between the selected user and product is no better than a random alignment. Cosine similarity was calculated for the conditions of all four encoders including TF-IDF and three LLMs for the purpose of performance comparison.

After cosine similarity, all the products were ranked in descending order for each user according to their similarity scores. This order would then be tested against the expert-labelled ratings, which was organized in descending order too. We used the metrics of mean average precision (MAP) to test how much the similarity order aligned with the real rating order. Mean average precision (MAP) is a ranking-based method designed to evaluate the ranking quality of recommended items. It has strong advantages when the varying relevance across the ranked list of items as well as the importance of high-ranked items are highly valued (Jadon & Patil, 2024, p.14-17). In our case, since we expected the items ranked higher with their similarity scores are more relevant to the corresponding user and more important among all the results from our recommender system, MAP would be a suitable choice as the evaluative metrics for cosine similarity. The result of MAP is always 0 and 1. Scoring 0 for MAP indicates none of the items with high ratings having ranked high among recommended items in terms of their cosine similarity scores. From 0 to 1, there is an increasing amount of match between items with high ratings and items ranked high with their similarity scores. Scoring 1 indicates an ideal recommender system performance with perfect match.

After having attained the cosine similarity scores and MAP for all four encoders, we chose the best performing model based on its MAP and applied feature engineering by assigning additional weights to users who have satisfied the condition for age and location (A more detailed explanation of feature engineering in our research could be found in the featuring engineering section above). The weighted features would be encoded again and received a recalculation of cosine similarities and MAP. Afterwards, a comparison would be made between the average precision from original feature data and that from weighted feature data.

For the final step, in order to test whether any difference between MAPs from different encoders or between the same encoder with original features and that with weighted features is statistically significant, we conducted the Wilcoxon signed-rank test. The Wilcoxon signed-rank test is a nonparametric statistical test used to compare two related samples. It calculated the average precision value difference for each user across two conditions to see if these differences are symmetrically distributed around 0 (Taheri & Hesamian, 2013). Given the small sample size we have for user data and product data, the Wilcoxon signed-rank test is suitable here for its nonparametric nature and its minimal assumptions about the underlying data distribution. In the end, the result of the Wilcoxon signed-rank test was reflected in p value.

3.4.2. Method 2: Simple machine learning using LLM embeddings

In the second method we introduce an approach based on supervised learning. Instead of computing pairwise similarity scores, this method directly predicts the expert-rated relevancy scores for each participant (patient)-card (product) combination using only the participant's textual profile. These profiles were embedded using pre-trained SentenceTransformer models. Each profile was transformed into a fixed-length semantic embedding, meaning that each profile will be a numerical vector that captures the meaning of the participant's characteristics, such as background, interests, and language style, in a way that can be processed by machine learning models.

These embeddings formed the input for three separate multi-output regressors (listed below), trained to map each vector to its corresponding set of twenty card relevancy scores.

1. **Random Forest**: This learner constructs hundreds of decision trees, each grown on a bootstrap resample of the training data and split with a randomly selected subset of predictors. The dual randomness (across observations and across variable) decorrelates the individual trees so that their averaged (regression) or majority-vote (classification) prediction exhibits markedly lower variance than any single tree, while still capturing complex, nonlinear interactions. Out-of-bag samples left out of each bootstrap also supply an internal estimate of generalisation error and support permutation-based importance scores, making the model simultaneously accurate and diagnostically rich (Breiman, 2001; Liaw & Wiener, 2002)
2. **Ridge regression:** Ridge augments ordinary least squares with an L2 penalty that proportionally shrinks coefficient magnitudes toward zero. By adding this "ridge" term the estimator counteracts multicollinearity, stabilises the XTX matrix, and trades a small, controlled bias for a potentially large reduction in sampling variance, often yielding lower predictive error when predictors are highly correlated
3. **k-Nearest Neighbours:** A non-parametric, instance-based technique that defers modelling until prediction time: for a new case the algorithm locates the k training observations lying closest in feature space (under a chosen distance metric) and returns the average of their response values. Because it assumes no functional form, kNN can adapt to highly irregular response surfaces; yet its efficacy hinges on the choice of kkk, the metric, and the dimensionality of the data, with performance degrading as irrelevant or sparse features dilute neighbourhood structure.

Then we applied an 80/20 train-test split to ensure generalizability, training all regressors on the same embedded input and evaluating them on a shared subset of participants whose data was set aside during training and used only to assess how well the models predicted unseen relevancy scores (test set). The evaluation metrics we decided to use were MSE and R^2.

- **Mean-Squared Error (MSE)**: MSE averages the squared gaps between the model's predictions and the observed outcomes, so every miss is expressed in the response variable's own units but "squared." Squaring serves two purposes: (i) it prevents positive and negative residuals from cancelling and (ii) it magnifies large mistakes, giving the metric a built-in aversion to outliers. Because ordinary-least-squares regression is the maximum-likelihood estimator under Gaussian noise, minimising MSE is statistically coherent with that distributional assumption (Bishop, 2006)

- **R^2:** R^2 re-expresses model quality as a ratio. It reports the proportion of total variance in the dependent variable that the model succeeds in explaining, with 1 indicating a perfect fit and 0 meaning the model is no better than predicting the sample mean. Applied-regression texts emphasise R^2's communicative power: "how much of what we want to predict is actually accounted for?", and caution that the raw statistic rises mechanically as predictors are added, motivating adjusted versions for fair comparison (Draper & Smith, 1998)

The combination that achieves the *lowest* MSE and *highest* R^2 on unseen data is declared the best choice.

3.4.3. Method 3: Fusion models

A fusion model combines multiple types of data (also called modalities) into one model to make predictions or recommendations. As no single data source is perfect, fusing different types of data gives the model a more complete picture of – in this case – both the person and the image on the card. Compared to the simple machine learning model, we are adding an extra layer of data as input for the model - so it basically takes the model of method 2 and adds a layer. In this study, we are using three data sources:

1. Tabular data about the demographics and preferences of the different participants (e.g., age, gender, location of birth, sports interests).
2. Semantic embeddings, created from the personal profile based on the data in the same file.
3. Card image embeddings created from the pictures on the cards used for the reminiscence therapy.

Fusing the different types of data allows the model to learn patterns across different modalities (Nourani, et al., 2023). As all the participants (30) received a relevancy score for every card (20) by an expert, we can use this to train the algorithm for it to recognize patterns. For instance, about how age and image content interact to affect preferences, visible through variation in the relevancy scores.

3.4.3.1 The visual model

As the tabular data is rather straightforward and the LLM models are already explained in previous sections, only the visual model will be explained in this paragraph. The visual models we are testing for our fusion models are all part of the group of multimodal vision-language models. These models are learnt to represent images and texts in the same embedding space. They enable powerful matching retrieval capabilities across modalities using contrastive learning. Contrastive learning teaches a model to pull similar inputs closer together and push dissimilar ones apart, instead of simply predicting a label (Xiao, et al., 2024). The model can make embeddings from visual data, by analysing the picture and comparing it with an enormous number of pictures it has been trained on. Moreover, this model also makes semantic embeddings, just like the LLM models we have used before (Kiros, Salakhutdinov, & Zemel, 2014).

In our case, we are only using the function of creating a visual embedding from a picture, as the semantic embedding is already created by an LLM model in previous steps. First, the model's image encoder converts every image into a 512-dimensional feature vector. These embeddings are then normalized and pooled. For every participant, twenty embeddings are created (one for each picture) – all these embeddings are combined into one visual feature. These visual factors are then concatenated with the tabular data and the semantic LLM embeddings: the full input to the fusion model.

There are different aspects of the fusion model that can be changed: the LLM model that is used to make the semantic embeddings, the visual model that is used to create an embedding from the picture, and the aggregation method used to pool the different image embeddings into one visual feature for every person.

- Like in other paragraphs, we have tested three different LLM models: MiniLM, Instructor-XL and All-mpnet-base-v2.
- For the visual models, we have tested Open AI's 'Clip' model and Facebook's DINOv-2 model. Both of them are state-of-the-art models, but they use a different approach. Where Clip is specialized in matching images with text, DINOv2 is specialized in understanding visual features only.
- As for the aggregation of the visual embeddings, we have tested two well-known methods: PCA and simple mean-pooling. Which are, respectively, the best methods when ratings vary and personalization matters and the best to get a strong baseline, especially with small datasets.

3.5. Step 5: Recommendation

The last step in the process is to use the now finished models to generate recommendations for new participants in the sports reminiscence therapy. This is the part that is eventually going to lower the labor-intensity of the matching process, as these recommendations for the matching of a new participant to the most relevant cards was done manually in the past - this would now be automated. Personal information about a new participant is provided, which is then used as input for the best performing model that is created in this study. The model uses what it has learned in the training phase and outputs relevancy scores for all the pictures that are included in the database that is being used in training. With these scores, a recommendation can be given on what cards would be the most relevant for the new participant. These results could potentially be checked by an expert - who has saved the time of doing this matching process in the first place and would now only have to check whether the output of the model is satisfactory.

4. Results

In this chapter, we will discuss all the outputs of the models - split up in the three different methods that we have used. Cosine similarity uses mean average precision (MAP) to assess the performance of the model. The other two methods are using test man squared error (MSE) and test R-squared

4.1. Cosine similarity:

Firstly, we conducted cosine similarity without additional feature engineering. Based on the cosine similarity calculated between users and products, the MAP was calculated for each encoder by testing its cosine similarity result against the labelled ratings. Among all four encoders, including 3 LLMs and one TF-IDF, the Instructor XL model received the highest mean average precision of 0.51. It signifies that, towards every user, around 51% of recommended items with high cosine similarity scores also possess high labelled ratings for the Instructor XL model. Having two out of three scoring higher mean average precision than the TF-IDF model, the LLMs performed generally better in recommending suitable items to users with dementia based on cosine similarity. However, the performance advantage of LLMs regarding TF-IDF is by no means definite given the poor mean average precision from MiniLm, which scored much lower than the result of TF-IDF. Wilcoxon signed-rank test was conducted to determine the statistical meaningfulness of any performance variation between different encoders. The results showed that the performance difference between every encoder was statistically significant with a p-value below .05.

Table 1: MAP results for different Encoders

Encoder	**TF-IDF**	**Instructor XI**	**MiniLM**	**all-mpnet-base-v2**
Mean Average Precision (based on cosine similarity)	0.34	0.51	0.30	0.44

Table 2: P values from the W test results between MAPs from different Encoders

Wilcoxon test (P value)	**TF-IDF**	**Instructor XL**	**MiniLM**	**all-mpnet-base-v2**
TF-IDF	1.0	<.01	.05	<.01
Instructor XI	<0.1	1.0	<.01	.01
MiniLM	.05	<.01	1.0	<.01
all-mpnet-base-v2	<.01	.01	<.01	1.0

Secondly, we have selected the Instructor XI encoder, which had the best performance in the previous step, applied different combination of additional weights with a grid search for feature engineering, recalculated the MAP with weighted data, and tested if any difference between the new MAP and the

MAP with the original feature data was statistically significant. The result showed that among all combinations of weight assignment in the grid search, an additional weight of 0.005 to users who were between 15 and 30 years old in the corresponding year of the product and an additional weight of 0.001 to users who resided in the same provincial location as that of the product helped achieve the best MAP of 0.49. However, this MAP with weighted features was 0.2 lower than that with original feature data and the difference was statistically significant based on the Wilcoxon signed-rank test.

Table 3: Results from weighted feature data using Instructor XI encoder

Encoder	**Weight on "age"**	**Weight on "location"**	**MAP (with weights)**	**MAP (without weights)**	**Wilcoxon test**
Instructor XI	0.005	0.001	0.49	0.51	<.01

Finally, based on the best performing result above, we used Instructor XI to encode six new user profiles provided by our external partner and calculated the cosine similarity with the encoded item description without assigning additional weights to the condition of age and location. We selected the five items with the highest similarity score to each new user and presented it in the table below as the recommended items. The difference between the top 5 recommended items in terms of their cosine similarity score for each user is only between 0 to 0.04 with the highest score around 0.75 and lowest around 0.72. Nonetheless, the validity of this prediction could not be tested until the rating of each item from these new users are received.

Table 4: Recommended items to new users

New users	Top-5 recommended items (with cosine similarity)				
New User 1	Kees Rijvers (0.75)	Cor van der Hart (0.74)	Jan Notermans (0.73)	Abe Lenstra (0.72)	Frans De Munck (0.72)
New User 2	Cor van der Hart (0.77)	Kees Rijvers (0.75)	Abe Lenstra (0.75)	Coen Moulijn (0.75)	Frans De Munck (0.74)
New User 3	Wim van Est (0.75)	Jan Nolten (0.75)	Coen Moulijn (0.74)	Jan Notermans (0.74)	Kees Rijvers (0.74)
New User 4	Ada Kok (0.76)	Kees Rijvers (0.73)	Fanny Blankers-Koen (0.73)	Coen Moulijn (0.73)	Cor van der Hart (0.72)
New User 5	Sjoukje Dijkstra (0.73)	Cor van der Hart (0.72)	Anton Geesink (0.71)	Fanny Blankers-Koen (0.71)	Ada Kok (0.71)
New User 6	Coen Moulijn (0.69)	Cor van der Hart (0.66)	Jan Notermans (0.66)	Ada Kok (0.66)	Frans De Munck (0.66)

4.2. Simple machine learning

As mentioned before, the next two sections will be using different metrics than cosine similarity did, so firstly we will explain these. The metrics we are using for this evaluation - test MSE and test R-squared - reflect the real-world performance, if the model were to be used with new data. The test MSE tells us the error on the test data: lower is better. The test R-squared tells us the proportion of variance explained on the test data: a high score means good performance on new data and that the model is able to capture a big portion of the variance that is visible in the relevancy scores.

The evaluation process orders the fifteen encoder–regressor pairings (Table 4) in a clear order. TF-IDF features combined with a Random-Forest regressor dominate the field, achieving a test-set mean-squared error of 0.899 and an R^2 of 0.871, well ahead of any rival. The best alternative comes from the large-language-model camp: MiniLM sentence embeddings fed to a ridge regressor post an MSE of 2.864 and an R^2 of 0.624. Feeding those same embeddings to a Random Forest slips slightly to an MSE of 3.115 and an R^2 of 0.585. Every other MiniLM, MPNet or Instructor-XL combination offers only modest improvements, and all k-nearest-neighbour models perform weakly, registering mean-squared errors above 5.8 and R^2 scores under 0.26. One-hot encoding consistently ends at the bottom; even with a Random Forest the error exceeds 6.0 and the share of variance explained (R^2) never rises above 0.24, confirming that plain indicator columns capture too little of the data's structure. In short, rich term-frequency information interpreted by an ensemble of decision trees gives the most dependable predictions, dense transformer embeddings work reasonably well when matched with a shrinkage-based linear model, and distance-based methods falter across all feature spaces.

Table 5: Results for text encoders & Large Language Models (Test MSE & Test R^2)

Test MSE	**One-hot encoding**	**TF-IDF embeddings**	**mpnet-base-v2**	**MiniLM-L12-v2**	**Instructor-XL**
Random Forest	6.029	0.899	5.251	3.115	4.956
Ridge Regression	6.089	7.210	6.047	2.864	7.273
KNN	13.838	7.703	6.415	5.829	5.802

Test R^2	**One-hot encoding**	**TF-IDF embeddings**	**mpnet-base-v2**	**MiniLM-L12-v2**	**Instructor-XL**
Random Forest	0.235	0.871	0.330	0.585	0.362
Ridge Regression	0.229	0.085	6.230	0.624	0.079
KNN	-0.740	0.012	0.171	0.258	0.261

Table 6: Top Recommendations for MiniLM-L12-v2 embeddings coupled with a Ridge regressor

New users	**Top-5 recommendation** **(predicted score trained on the already expert-given matching score ranging from 1 to 10)**				
New User 1	FBK (10.00)	Ada Kok (9.76)	Anton Geesink (8.96)	Wim Van Est (4.95)	Kees Rijvers (3.69)
New User 2	Anton Geesink (10.00)	Ada Kok (7.27)	Kees Rijvers (6.00)	Faas Wilkes (5.44)	Jaan Notermans (5.41)
New User 3	Ada Kok (10.00)	FBK (9.82)	Anton Geesink (9.42)	Kees Rijvers (6.90)	Jan Notermans (6.79)
New User 4	Wim Van Est (10.00)	Ada Kok (9.07)	Jan Nolten (8.31)	Jef Lahaye (8.31)	Sjoukje Dijkstra (7.06)
New User 5	Kees Rijvers (10.00)	Faas Wilkes (9.07)	Jan Notermans (9.02)	Cor Van Der Hart (9.02)	Fortuna 54 (9.02)
New User 6	Wim Van Est (10.00)	Ada Kok (9.10)	Jan Nolten (8.55)	Jef Lahaye (8.55)	Sjoukje Dijkstra (7.36)

4.2.1. Comments on the ML results

Because the TF-IDF + Random-Forest result raises the flag for overfitting (further analyzed in discussion section) , **MiniLM-L12-v2 embeddings coupled with a Ridge regressor** take the crown as the most robust, generalisable pairing for the machine learning approach.

4.3. Fusion models

The fusion model continues where the method above stops and builds on top of it by adding a visual layer. There are three different aspects in which the fusion model can be altered: the LLM used to make the semantic embedding, the visual model used to make the visual embedding, and the aggregation method to combine all the visual embeddings made for different pictures. Below, all the results are shown when using different models/approaches for these three aspects of the fusion model.

Table 7: Results for different LLMs

Method (using PCA as agg. method + Clip as vis. model)	Train MSE	Test MSE	Train R^2	Test R^2
MiniLm	1.136	3.283	0.874	0.565
Instructor-XL	1.265	4.350	0.860	0.447
All-mpnet-base-v2	1.252	4.417	0.861	0.430

Three different LLM models have been tested, keeping everything else constant - the MiniLM model is performing better than the other two, with an R-squared value of more than 0.1 higher than the other two models.

Table 8: Results for different visual models

Model (using PCA + MiniLM-L12 as LLM)	Train MSE	Test MSE	Train R^2	Test R^2
Open AI's Clip	1.136	3.283	0.874	0.565
Facebook's Dinov2	1.142	3.324	0.874	0.557

Two visual models have been tested, keeping everything else constant - Open AI's model is the best performer, though the results are very close together.

Table 9: Results for different aggregation methods

Method (using MiniLM as LLM + Clip as visual model)	Train MSE	Test MSE	Train R^2	Test R^2
PCA	1.136	3.283	0.874	0.565
Mean-pooling	1.156	3.393	0.872	0.550

Two common methods to aggregate embeddings have been tested, keeping everything else the same. There was not a big difference visible between the two, though PCA performed slightly better. Now we know that using MiniLM to create the semantic embeddings, using Clip to create the visual embeddings, and using PCA to pool these visual embeddings, is the combination that leads to the fusion model with the best results for this task.

Table 10: Results compared to simple machine learning

Approach	Train MSE	Test MSE	Train R^2	Test R^2
Fusion (Tabular + LLM + Vision)	1.136	3.283	0.874	0.565
Tabular + TF-IDF + Vision	0.147	0.758	0.981	0.890
Tabular + LLM	1.124	2.908	0.876	0.616
Tabular + TF-IDF	0.154	0.775	0.981	0.888

When compared to the same approach, without adding the extra visual layer, or swapping the LLM for a TF-IDF encoder, it becomes clear that the addition of the visual model worsens the performance of the model. Using TF-IDF instead of the LLM model to create semantic embeddings is a significant improvement as well, which could indicate that using the LLM for this task is not ideal - or that TF-IDF is overfitting. As the results are significantly worse than for other methods we used, we have chosen not to use this model to create recommendations for new users.

5. Conclusion

We will first provide sub conclusions for the different methods we have used, before answering the research question.

5.1. Cosine similarity

In general, encoders used in our recommender system based on cosine similarity function captured the relevancy between users and items with low to immediate precision. Instructor XI performed the best among all encoders with 0.51 MAP and MiniLm performed the worst with 0.30 MAP. The reason behind this could be the parameter size difference between them. Instructor XI has more than 1 billion parameters, which renders it higher potential to capture the more nuanced, contextual and semantic meanings than MiniLm or all-mpnet-base-v2, which have around 30 millions and 100 millions parameters respectively. Besides, LLMs performed generally better than TF-IDF in recommending relevant items to users. This could be explained by the fact that TF-IDF emphasizes merely on the lexical level in calculation whereas LLMs tend to concentrate on analyzing the information on a contextual level. The difference between model performance measured by MAP was statistically significant across every pair of model comparisons, which offers more confidence in choosing a larger, more comprehensive LLM with more parameters as the encoder for a recommender system for people with dementia based on the ranking function of cosine similarity.

Furthermore, assigning additional weights to users who have satisfied the age and location conditions defined and suggested by expert knowledge and corresponding theories did not lead to a better MAP for Instructor XI model. Instead, it decreased the MAP with statistical significance. This result is opposite to our expectation. One possible explanation concerning age condition is that the time range of age 15 to 30 is so broad that it resulted in granting the majority of users in the dataset an additional weight and thus heavily tampered the distinguishing effect from weight assignment. A possible solution for this could be narrowing it down to a much smaller age range or even to a specific year that was most important to the corresponding user. Another possible explanation concerning age condition is that not all items have their corresponding time specifically known and stated in the item description, which resulted in some potential time match between users and items being overlooked in this process. A possible solution is to ascertain every item has its most relevant year or time range specifically defined and stated in the item description. The same logic applied to the additional weights concerning the location condition as well. Using province as the spatial range for users could be too broad and not all items have their most relevant location specifically stated in the description. Overall, in order to have additional weight assignments perform better, future steps regarding further fine tuning the time and location range for each user and item and a clear statement of such information in descriptions are highly recommended.

Lastly, we have made recommendations of top-5 relevant products to new users provided by the external partner using cosine similarity with Instructor XI encoder and without additional weights. This result could be tested against labelled ratings yet to be acquired or through field study in future situations.

5.2. Textual Machine learning & Fusion Model

Because both the purely textual and the text-plus-vision “fusion” approach rely on supervised learning, we evaluated them side-by-side to see whether more complex input translates into higher predictive power.

We began with the initial guess that large-language-model (LLM) sentence embeddings (MiniLM, MPNet, Instructor-XL) would outperform simpler TF-IDF vectors, since the transformers encode far more semantic structure. However, the output numbers told a slightly different story: TF-IDF paired with a Random-Forest regressor performed the best with an MSE of 0.899 and an R^2 of 0.871. Having a closer look at the results we can see that the random forest overperforms compared to the other regressors (Ridge Regression) signalling likely overfitting. Setting that result aside, the most reliable text-only model was MiniLM-L12-v2 embeddings with Ridge regression (MSE 2.864 / R^2 0.624); a combination that balances rich representations with a variance-controlling linear learner. All other LLM-plus-Random-Forest or KNN variants lagged behind ($R^2 \leq 0.59$), and one-hot encoding never rose above 0.24, confirming that indicator columns capture too little structure.

Adding a visual layer did not close that gap. The best fusion model (MiniLM text embeddings + CLIP visual embeddings aggregated by PCA) reached an MSE of 3.283 and an R^2 of 0.565; a (moderately) good but clearly lower score than the MiniLM-Ridge text model without the fusion layer. Replacing MiniLM with TF-IDF gave the fusion model performance a boost, but the same warning sign appeared: the difference between train and test performance was very large, pointing to overfitting. In short, adding the visual layer made the model more complex without giving matching benefits, and the LLM embeddings—though in principle richer—did not reliably outperform a well-regularised linear model that uses TF-IDF features. These findings make MiniLM-L12-v2 with Ridge regression the most robust choice and show that the fusion approach will need tighter regularisation or a larger data set before it can reach its full potential.

5.3. Answering the research question

So, to answer the research question: To accurately match a personal profile of people with dementia to the most relevant sports image, we have tried both memory-based approach (cosine similarity) and model-based approach (simple textual machine learning & machine learning with fusion models) with four different encoders (3 LLMs & TF-IDF). For cosine similarity, choosing a larger LLM with more comprehensive parameters like Instructor XL performed better in our recommendation task. Moreover, assigning additional weights based on expert-defined age and location conditions demonstrated a negative impact to the performance, but it seems to bea more likely caused by dataset incompleteness and it is recommended to conduct this test again with more complete and larger datasets in future research. For the two model-based approaches, on the other hand, choosing a smaller LLM such as MiniLm gave better performance. Compared to TF-IDF, it balanced better effectiveness of the learning process and the prevention of potential overfitting. Besides, compared to the visual information applied in the fusion model, using only textual information regarding personal profile and item description achieves higher accuracy in our case.

6. Discussion

In the discussion section, we will discuss things that are still worth mentioning about the process or the data we have used. This includes some limitations, and a number of recommendations that can be made for further research.

One of the most significant challenges encountered during this research project was the nature of the dataset. The first issue was its size. As previously mentioned, fewer than 20 candidates participated in the study, which made training our models harder and likely caused overfitting. With such a small sample, our models may have learned patterns that were specific to the training data rather than generalizable trends, reducing their performance on new unseen data.

The second issue was the subjectivity involved in the matching scores between the candidates and the products. In our case, the scores were assigned by a single expert, which introduces the risk of bias. Training models on scores that were assigned in this manner may result in recommendations that reflect personal preferences rather than broadly applicable patterns.

A potential solution to both of the above problems is to collaborate with more organizations, such as caregiving institutions or relatives of the participants. Expanding the network of collaborators would naturally increase the size of the dataset by involving more individuals with dementia. In addition, having multiple caregivers or experts provide relevancy scores would allow for averaging their evaluations, resulting in more balanced and representative targets for model training. With the expansion of the data, we believe that we could also use the full potential of our models, especially while using LLMs, since transformer-based sentence encoders reach their full strength only when they are exposed to a rich, domain-specific corpus. A larger and more varied set of biographies and pictures (+ descriptions) would allow us to fine-tune the pre-trained language models on Dutch sporting narratives and care-home vocabulary, giving the embeddings a more meaningful sense of regional terms, idioms, and stylistic cues.

Something else that is worth mentioning is that we attended an event hosted by “Voetbal Herinneringen” where we observed that individuals with dementia appeared more responsive to memory triggers in social environments — such as group gatherings and conversations — rather than to isolated stimuli like static products or cards. The participants and supervisors told us the fact that the participants were able to meet friends, and other people was in itself more valuable than the product they were using (reminiscence prompts like photographs). This suggests that while the use of memory-triggering cards is a valuable and creative approach, it may lose some of its effect when they are not used in a group setting.

This observation is further supported by the characteristics of the participants themselves. Most, if not all, were elderly individuals with challenges in systematic communication, which makes consistent participation in such research more difficult. Their involvement often requires the support of relatives or professional caregivers to help with engagement and interaction.

Further notations that have to be made in the discussion section is the use of generative AI in order to handle the variety of errors while coding, and to get suggestions for literature. Although it has been used, this was in no way of any extensive form, emphasizing the authenticity of our work.

Ethical considerations were given high priority throughout the process. Because of the sensitive nature of dealing with individuals who have been affected by dementia, particular care was taken so that all personal details were kept as anonymous as possible. Participants' privacy and dignity are not only required by law, but also by ethics. No information that could potentially reveal any individual's identity was retained or passed on, and all data sets were handled in accordance with GDPR regulations. Further, informed consent was also obtained when the data were collected — either directly from the participants when possible or through legal guardians or caregivers. Ethical research in this context also involves considering the cognitive limitations of the participants and ensuring that any form of interaction is non-intrusive, respectful, and designed to generate well-being and not stress or confusion.

For further research, it would be recommendable to use a bigger dataset and test how these models perform. This should then be linked to a database that is bigger than just the twenty cards that were available for us right now. If this is done, the system can actually start being used in real-life, automating the matching process and therefore reducing the work that needs to be done by the therapist. Ideally, the final result should be tested in different settings: a group setting and a setting in which the person receives the treatment individually, to research which of these has the biggest positive effect on the participants of the reminiscence therapy. Furthermore, it would be a good idea to look at when we consider the therapy as successful - how are we evaluating whether a picture “works”? Because we have been using a ‘relevancy score’ looking at e.g., age, gender, and place of birth, but the fact that a picture is relevant to a person does not necessarily mean it will trigger emotions or be meaningful to a person. It would be interesting to research which one of the 5/10/20 most relevant pictures actually trigger a response that we are looking for and make the participants benefit most from the therapy. Because that will of course always be the main goal - making the positive effect of (sports) reminiscence therapy on the well-being of the participants as meaningful as possible.

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8. Appendix - Visit Voetbal Herinneringen Tilburg

Voetbal Herinneringen (football memories) is a company that aims to help elderly with dementia by using sports reminiscence. The participants are gathering at a football club that binds them - in this case Willem II in Tilburg. In contrast to Sport and Memory, Voetbal Herinneringen makes the participants come to them instead of the other way around. The evaluation of “success” is simply done by looking at whether participants are returning to future meetings.

These gatherings take the same amount of time as a football match: 2 “halves” of 45 minutes and a break of 15 minutes. During the two halves, people are free to talk about whatever they want, but a lot of materials about the football club in question are provided. These materials could steer the conversation, just like the location and the fact that it is a “Willem II gathering”, but participants are free to talk about whatever they want. For roughly every two participants, there is a volunteer on site. These people help with organising the sessions and try to keep the conversations flowing by asking the participants questions about their personal life and the football club. The 15-minute break is the part that is directly led by the organisation and changes every week: one week, a therapist hosts a session on maintaining motor skills, the other week a former first team player is being interviewed by one of the volunteers.

When asking the participants about what brings them to the meetings and makes them come back, most mention that they like seeing the other participants and getting to know new people. As many live in care homes and/or struggle with loneliness, having gatherings with peers is very valuable. Even though a lot of things are discussed that do not have to do with football, football is usually what binds these people in the first place. As they start from a shared interest, it is easy for the participants to connect with each other. Also, the fact that they leave their homes and make a trip to a place they have fond memories of, is very important to them. The football related materials mainly serve a role as conversation starters but are also used with the participants that suffer from severe dementia to help them remember things.